

Data Quality Issues & Fixes

During the data preparation phase, several data quality issues were identified and resolved to ensure accurate analysis:

1. Duplicate Records

- Issue: Duplicate rows were present in both datasets
 - Fix: Removed duplicates using full-row comparison in Power Query
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2. Invalid Text Values (NaN, NAN, Blanks)

- Issue: Text columns contained inconsistent null representations
 - Fix: Replaced all such values with proper nulls, then filled with "Unknown" where necessary
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3. Missing Values

- Issue: Missing values in numeric and categorical fields
 - Fix:
 - Numeric columns → replaced with 0
 - Text columns → replaced with "Unknown"
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4. Incorrect Data Types

- Issue: Date columns contained errors and inconsistent formats
 - Fix: Cleaned invalid entries and converted all date fields to proper date format
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5. Negative Values in Metrics

- Issue: Negative values found in clicks, impressions, and spend
 - Fix: Converted to absolute values and added validation flags
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6. Decimal Values in Count Columns

- Issue: Clicks and impressions contained decimal values
- Fix: Converted to whole numbers using rounding

7. Recalculation of Metrics

- Issue: Derived metrics like CTR were unreliable or missing
- Fix: Recalculated CTR, CPC, CPM, and ROI using base columns

8. Inconsistent Text Formatting

- Issue: Mixed-case values in platform and region fields
- Fix: Standardized formatting using consistent capitalization

9. No Direct Relationship Between Tables

- Issue: Campaign and sales tables lacked a common key
- Fix: Used date dimension as a bridge for analysis

Conclusion

All transformations were performed within Power Query to ensure a reproducible and transparent data cleaning pipeline. The final dataset is structured, consistent, and suitable for analytical reporting.